ZM002 Artificial Intelligence

Assignment for PI7 assignments.

During this minor, you will be doing three assignments for PI7. In these assignments, you will dive deeper into applying AI techniques for a given context. More details about each individual assignment can be found below. The assignments can be made individually or in pair. However, groups of 3 or more students are not allowed. Each student, also if working in a pair, has to hand in the assignments individually on moodle.

Important! The report and code are graded, but importantly you also have to provide a short video **(15 min)** where you explain your report and code. So this means that for all three assignments you also have to hand in a max 15 minutes video with an explanation of your own work.

The three assignments are:

1. Evolutionary algorithms
2. Neural networks
3. Auto-encoders and variational auto-encoders

On the following pages, you will find the details of each assignment.

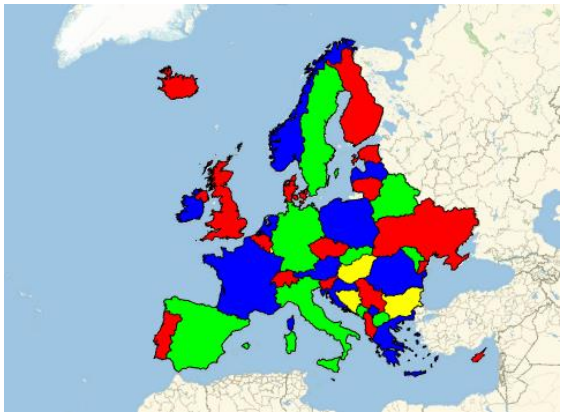
Part 1 – Evolutionary algorithms

Deadline

Submit the results of this assignment via Moodle, no later than **Monday of week 5** of the block period, before 11.59 PM (23.59h).

Goal

In the first assignment you will implement a genetic algorithm. Your goal is to colour the map of Europe, so no two bordering countries have the same colour. An example solution can be found in the picture below.



You are expected to choose a fitting genetic representation and research possible sub-algorithms for recombination, mutation, parent selection and survivor selection. Based on your research, you will implement the genetic algorithm using the found sub-algorithms. It is important that you substantiate the chosen sub-algorithms for your implementation. You are allowed to make use of the existing implementations of genetic algorithms from the discussion groups to use as a starting point.

After implementing the genetic algorithm you will evaluate its performance using different hyperparameters and/or different sub-algorithms. You are expected to interpret the results and discuss the best combination.

Results

- **A research report** detailing the results of the literature review of the genetic algorithms sub-algorithms (and alternatives). The report contains the used references and an explanation of the workings of the methods as well as a theoretical comparison of the pros and cons of the methods.
- **A realization** of the chosen options:
    - A Python code base or notebook with the implementation of the algorithms.
    - An explanation of the code using in-code comments.
- **An evaluation report** containing the results of, and comparison of, the output of the algorithms. Here it is important to show quantitatively how the different choices made effect the results of the genetic algorithms. This report should end with a conclusion stating what the best combination is and why.

Part 2 – Neural networks

Deadline

Submit the results of this assignment via Moodle, no later than **Monday of week 7** of the block period, before 11.59 PM (23.59h).

Goal

In this assignment you will design and implement a deep neural network for the following context:

"The company SLIFTS (Smart Lifts in Floor Transition Systems) wants to expand its elevator capabilities to handle spoken commands. In the aftermath of the global 2020 COVID pandemic, the company has noted a sharp decline in the number of passengers that use their elevators. Marketing research has shown that people are hesitant to touch physical buttons in the elevator. As one user noted "this up-button looks really yucky, I can almost see the germs crawling on it!". The situation is extremely serious and people are even doing previously unthinkable things like taking the stairs, which has to be prevented in all cases. To resolve this problem, SLIFTS has hired Zuyd University of Applied Sciences to research and develop elevators with voice command capabilities.

You are asked to research how neural networks can be employed to enable voice commands. As a proof of concept, the mini speech commands dataset can be used as a stand-in for the actual data (https://www.kaggle.com/datasets/antfilatov/mini-speech-commands). SLIFTS asks your advice on how to pre-process the data and what kind of NN architecture is suitable for the task of recognising a fixed number of speech commands. Of course, an artefact demonstrating the capabilities and an evaluation of the system are also required."

It is expected that you research different neural network architectures and preprocessing possibilities. Based on your research, you give a substantiated advice to SLIFTS, implement the neural network and evaluate its performance.

Results

- **An advisory report** containing an analysis of the given problem description, the results of the research for preprocessing and neural network architecture, a proposal for a possible solution including alternatives, and a final advice.
- **A realization** of the solution given in the advice:
    - A Python code base or notebook with the implementation of the DNN.
    - An explanation of the code using in-code comments.
- **An evaluation report** containing the performance metrics for the DNN, which are interpreted and discussed.

Part 3 – Auto-encoders and variational auto-encoders

Deadline

Submit the results of this assignment via Moodle, no later than **Monday of week 9** of the block period, before 11.59 PM (23.59h).

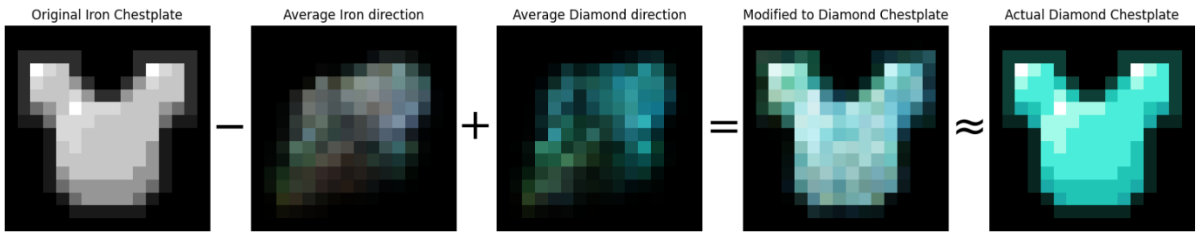
Goal

In this assignment, you will study Auto-Encoders (AEs) and Variational Auto-Encoders (VAEs) in the context of generating Minecraft item textures. You are provided with a dataset containing Minecraft item textures (available on Moodle) and are required to train both an AE and a VAE on this dataset.

The objective is to compare the two models along two key dimensions: reconstruction quality and latent space structure.

Reconstruction quality refers to how accurately each model can reproduce input textures after the encoding and decoding process.

Latent space structure is evaluated through latent space arithmetic, where vector operations in the latent space are used to generate new textures with modified semantic attributes. As an illustrative example, concepts such as an iron chestplate, the material iron (based on averaging all iron items) and the material diamond (based on averaging all diamond items) can be encoded by a latent vector representation. A semantic transformation can then be performed using a vector operation such as:



You are expected to analyze and discuss the performance of both models based on the results of your experiments, and to support your findings with relevant literature explaining why each model performs well or poorly in the examined aspects, including research towards anisotropy and isotropy.

Results

- **A research report** detailing the results of the literature review on AE and VAE. You are expected to specifically focus on which model should perform well for which key dimension and why.
- **A realization** in which:
    - It is shown that you successfully trained the AE and VAE.
    - One or more latent space arithmetic experiments are performed, such as the example described above, to illustrate semantic transformations in the latent space.
    - An explanation of the code using in-code comments.
- **An evaluation report** containing the results of, and comparison of, the output of both AE and VAE. Here it is important to show both **qualitatively** and **quantitatively** how well both models generate (new) Minecraft textures and how much compression each model allows before reconstruction fails.